CA-3 · FOUNDATION OF AGENTIC AI

2026

TY AI-ML B · SIT

DRILLING

Agentic AI

DAILY DRILLING REPORT

OIL AND GAS

NPT DETECTI



Reading the rig: finding lost drilling time in daily reports with a LangGraph multi-agent system.

Rahul Narkhede (24070126146) and Ranjan Verma (24070126149)

EVERY HOUR OF RIG TIME





40-50% of capex

Drilling and completing wells takes 40–50% of an offshore operator's capital spend. Operators that track lost time closely have cut NPT from 16% to 9% (Source 9).

~$400k a day

A drillship costs close to $400,000 a day in late 2025. Every hour the well does not move forward burns about $16,000 of rig time (Source 10).

Agents are arriving

In 2025 ADNOC gave AIQ a $340M contract to roll out ENERGYai agents across 28+ fields. The industry is now asking how far agents can be trusted (Sources 11, 4).

INDUSTRY CONTEXT



THE REPORT

Every rig sends a Daily Drilling Report (DDR) each morning: an hour by hour time log, depth drilled, bit and mud data, weather, crew and the cost of the day. The IADC has been reworking the DDR so activities are tracked in more detail (Source 2).

06:00–08:30 Drilled 12¼ in hole, 2,450 to 2,610 m
08:30–14:00 Pulled out of hole, MWD tool stopped sending data
14:00–16:00 Waited on weather, wind above crane limit
Most of it is free text, so lost hours hide in sentences.

WHERE HOURS GO

74% from two causes. In a study of wells on the Lobo trend, stuck pipe and lost circulation alone made up about 74% of all non-productive time (Source 1).

NPT means hours the rig is paid for but the well does not move forward.



Other common causes are tool failure, waiting on weather, waiting on equipment and well control (Source 3).

Today engineers spot these by eye in free text, so small losses are never added up and the same failure comes back on the next well.

NON-PRODUCTIVE TIME



THE PROBLEM

How can an engineer get a reliable account of lost time from every daily drilling report without reading each one line by line?

Reports arrive every day from many rigs, written as free text by the crew.

Our answer: a team of AI agents reads each report, measures the lost hours, finds the likely cause from past events and checks its own work.

A short evidence backed summary then goes to the drilling engineer, who approves or rejects it.

PROBLEM STATEMENT

APPROACH

MANUAL TODAY
An engineer reads each report by eye. Lost hours are not totalled, labels differ from person to person and lessons from past wells are not reused.

PLAIN CHATBOT
One prompt, one answer. The model guesses the sums, cannot check itself and gives no proof for what it says. Easy to demo, hard to trust.

OUR AGENTIC SYSTEM
Plans the work, uses Python for exact sums, compares with the well plan, checks itself with a Critic and waits for the engineer to approve.



THREE WAYS TO READ A REPORT



SIX ROLES, ONE STATE

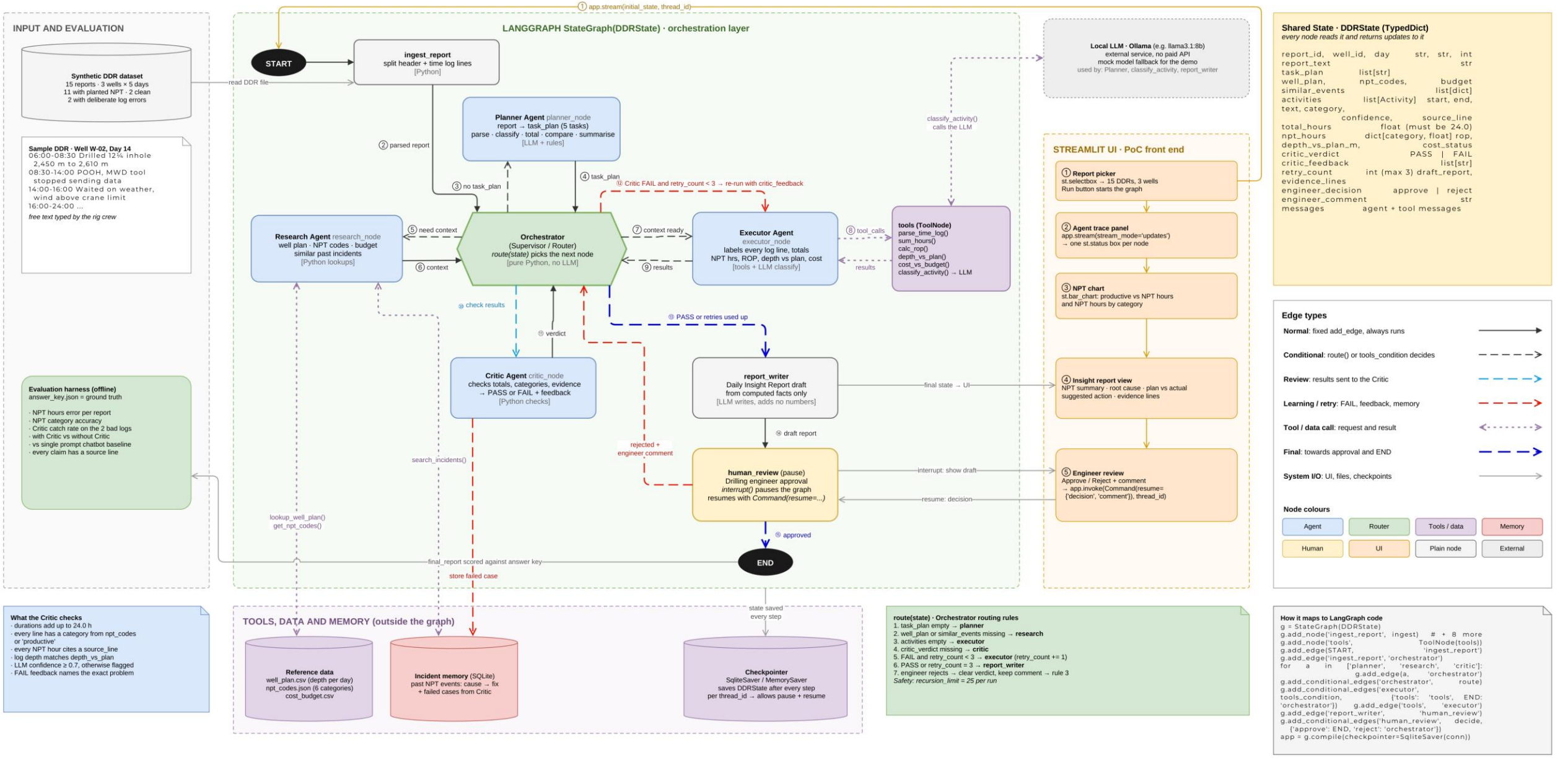
ORCHESTRATOR reads the shared state and decides who works next.
PLANNER turns one report into tasks: parse, classify, total, compare and summarise.
RESEARCH pulls the well plan, NPT codes, daily budget and similar past incidents.

EXECUTOR calls Python tools to label each log line and total NPT hours, ROP and cost.
CRITIC checks the log adds to 24 h and every NPT hour has a category and a source line.
ENGINEER is the human who approves or rejects with a comment.

AGENTS

Agentic AI System for Daily Drilling Report (DDR) Analysis and NPT Detection

LangGraph architecture: Orchestrator hub and spoke, Critic retry loop, ToolNode for exact calculations, engineer approval (human in the loop)



EDGE TYPES

NORMAL: a fixed step that always runs, like START to ingest or an agent handing back to the Orchestrator.
CONDITIONAL: route() or tools_condition picks the next node from the state.

REVIEW: finished results go to the Critic for checking.
LEARNING / RETRY: a Critic FAIL sends work back with feedback, up to 3 times, and failed cases are saved to memory.



TOOL / DATA: agents call Python tools, reference data and the incident memory.
FINAL: a Critic PASS or used up retries goes to the engineer, then to END.

SYSTEM I/O: the Streamlit UI, DDR files and checkpoints that let the graph pause and resume (Source 5).

HOW THE GRAPH MOVES



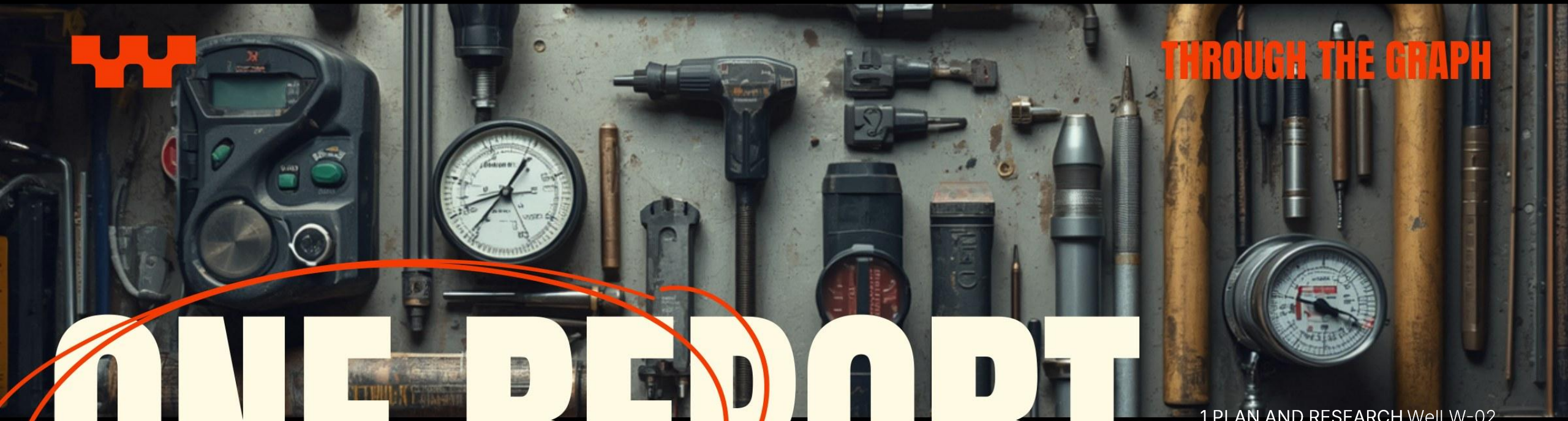
LLM + TOOLS

PYTHON

does the maths

THE MODEL READS: a local Ollama model (qwen2.5:3b) only labels each log line with an NPT category and a confidence score. It does not write the numbers: in testing it swapped which hours belonged to which category. A mock classifier is the fallback and is always shown as mock.

PYTHON COUNTS: tools parse the time log, sum the hours, work out ROP and compare cost with budget, so every number is exact and repeatable. Stack: Python, LangGraph, LangChain, Ollama, Streamlit (Sources 5 to 8).



ONE REPORT

THROUGH THE GRAPH

1 PLAN AND RESEARCH Well W-02, Day 14. The Planner makes 9 tasks. Research finds 2,650 m planned against 2,610 m actual and two past MWD failures on this field.

2 EXECUTE AND CHECK Tools label 5.5 h of tool failure. The Critic sees 23.5 h, names the 20:30-21:00 gap and sends it back. The retry finds it in REMARKS: 24.0 h, PASS.

3 ENGINEER DECIDES The model also called 'slip and cut' (3 h) NPT. The engineer rejects with 'line 7 is planned', the graph re-runs: 8.0 h NPT, approved.

DUMMY DATA

DATASET

known answers

15 REPORTS, 3 WELLS, 5 DAYS EACH: 11 have planted NPT events (tool failure, waiting on weather, stuck pipe, lost circulation), 2 are clean days and 2 carry deliberate errors, such as a time log that adds to 23.5 hours.

REFERENCE FILES: well plan depth per day, NPT category list and daily cost budget, plus a memory table of past incidents with cause and fix. We planted every event ourselves, so we know the right answer for each report.



STREAMLIT UI

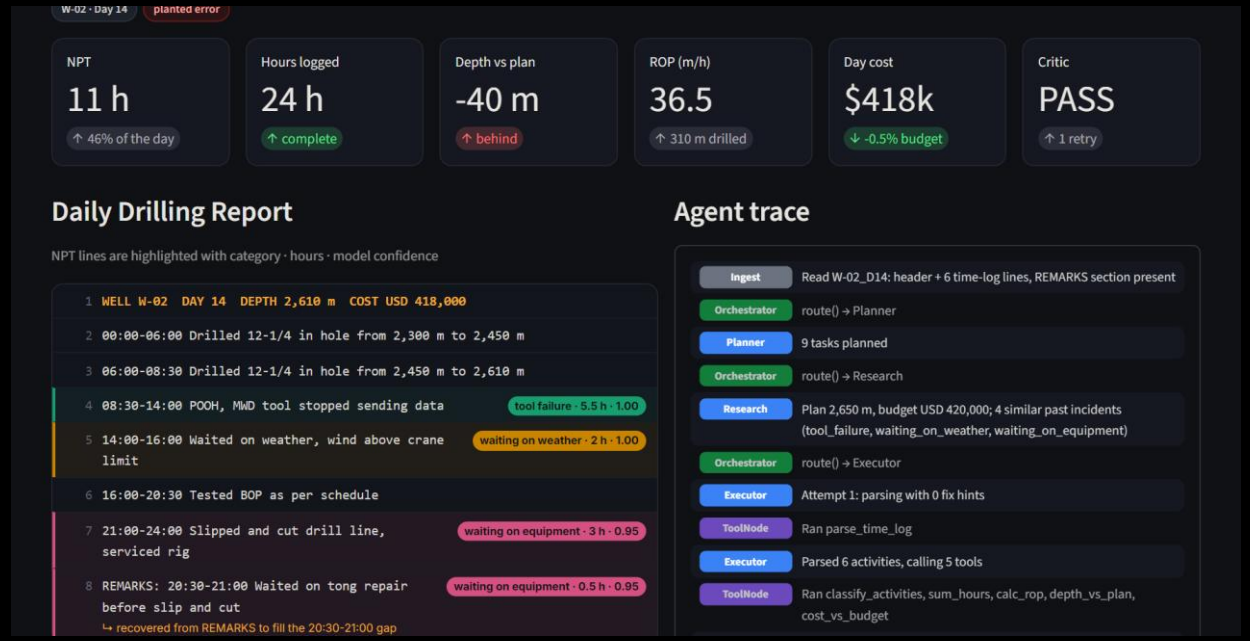
1 Pick a report and see the raw text
2 Watch each agent run with its output shown live
3 See productive vs lost hours and NPT by category in a chart
4 Read the insight report with source lines
5 The engineer clicks Approve or Reject

TEST CASES: a clean report, a tool failure report and a report where the Critic catches an error (Source 7).

DEMO

W-02 · DAY 14, REAL RUN

THE APP, LIVE





Run: each NPT line labelled with category · hours · model confidence; the 20:30-21:00 line recovered from REMARKS after the Critic's FAIL; the live agent trace.

After review: the engineer rejected one mislabel (line 7, slip and cut), the graph re-ran: 8.0 h NPT, approved.

Try it: rig-sight.streamlit.app (hosted on Streamlit Community Cloud, runs on the mock classifier; the real model runs locally with Ollama)

METRICS

Every synthetic report has a known right answer, so we score the system on NPT hours error per report and category accuracy across all planted NPT events.

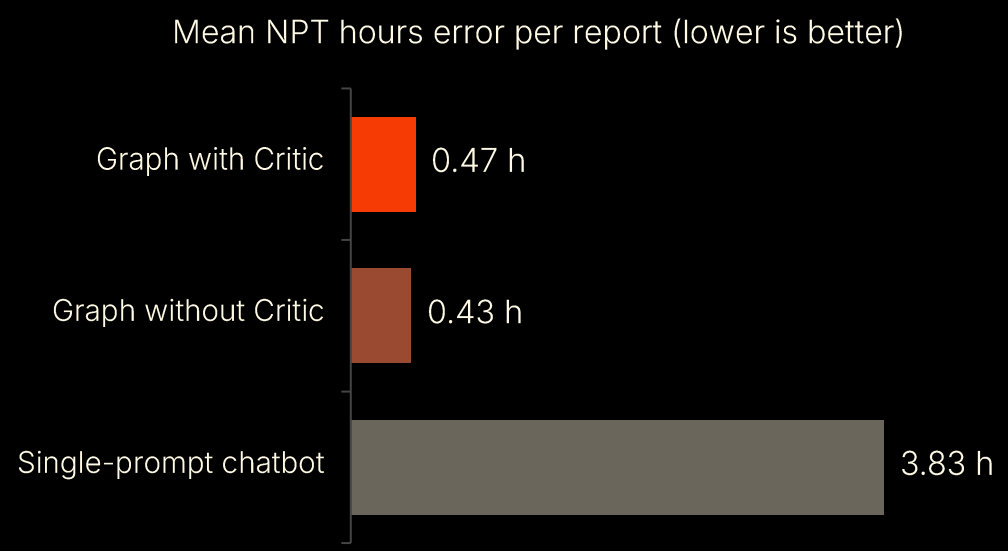
We also track the Critic catch rate on the 2 reports with planted errors, and run the same 15 reports without the Critic and through a single prompt chatbot. Every claim must point to a line in the report.

RESULTS (qwen2.5:3b, 15 reports):
NPT error 0.47 h per report for our graph vs 3.83 h for a chatbot on the same model. Next slide.



SAME MODEL, 15 REPORTS

RESULTS



	With Critic	No Critic	Chatbot
Category accuracy	100%	96.6%	76%
Critic catch rate	2 / 2	n/a	n/a
Logs adding to 24 h	15/15	13/15	1/15
Reports exactly right	13/15	13/15	4/15
Evidence rate	100%	100%	96%

What it shows: on the same model, splitting the job and letting Python count cuts the NPT error from 3.83 h to 0.47 h per report. The chatbot cannot even add the log up to 24 h. The Critic caught both planted errors before any human saw them.

Read it honestly: 'No Critic' has a slightly lower error (0.43 h) only because two mistakes cancel on W-02 Day 14. Our two remaining errors are planned work the model labels NPT with high confidence (slip and cut, waiting on cement): the engineer fixes each with one reject comment. Chatbot: latest run; separate runs on other days gave 3.77 h and 4.43 h. Mock fallback: 0.27 h. 48 automated tests pass.



LIMITS + NEXT

LIMITS: synthetic data, six NPT categories, one report format. The 3B model labels some planned work as NPT with high confidence. Not a certified engineering or safety tool.

TRUST: the industry is still deciding how far to trust agents with operational decisions, so the engineer keeps the final say (Source 4).

NEXT STEPS: live rig data feeds and many wells at once, learning from engineer comments over time and a morning alert to the drilling team.

LATER: MCP connectors so the agents can read company systems directly.

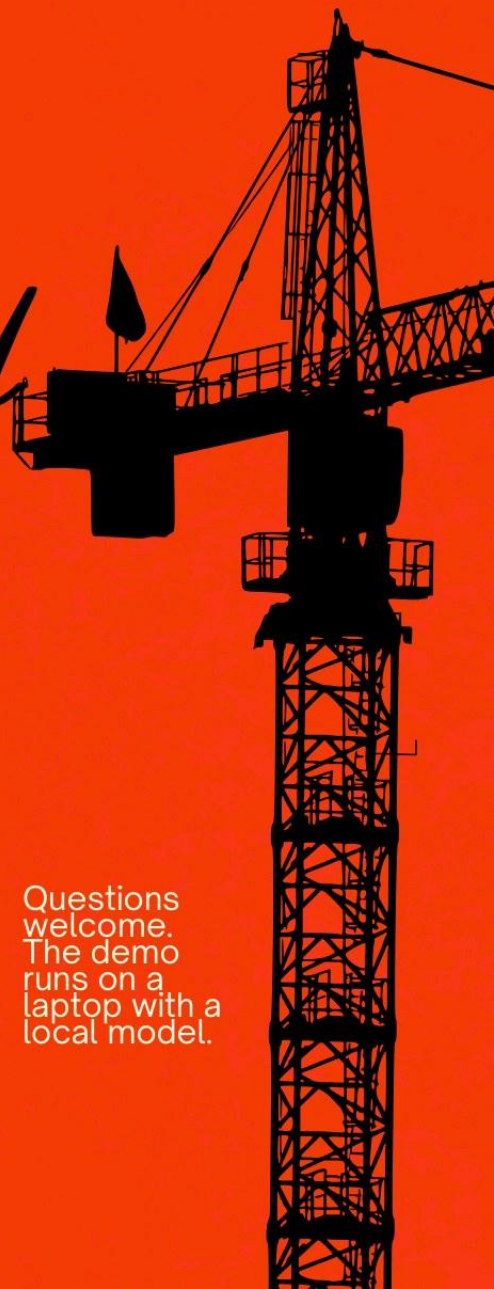
BEYOND CA-3

THANK YOU

computed, sourced, approved.

Every number computed by Python. Every flag sourced to a log line. Every report approved by an engineer.





Questions welcome. The demo runs on a laptop with a local model.

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